

Street Light Defect Detection Using ML and CV

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OBJECTIVE

The primary objective of this project is to develop a system that can automatically identify defective streetlights in both urban and rural areas using lightweight machine learning and computer vision techniques. The system aims to detect streetlights from captured visual data and determine their operational status by analyzing the light intensity on the road directly beneath them. This enables efficient detection of lights that are dim, flickering, or completely non-functional, while also supporting automated reporting of their locations for timely maintenance.

PROBLEM STATEMENT

Streetlights play a crucial role in ensuring road safety and visibility during nighttime. However, in many urban and rural regions, streetlights are not regularly inspected or maintained. As a result, faulty lights often go unnoticed for extended periods, leading to safety hazards and inefficient energy usage.

Manually checking each streetlight is both time-consuming and resource-intensive, especially in large areas. Additionally, existing systems often lack real-time reporting mechanisms and rely on public complaints or periodic manual inspections.

The core challenge lies in automatically identifying defective streetlights and pinpointing their exact location so that maintenance teams can take immediate action. To address this, the project proposes a vision-based approach where the operational status of a streetlight is inferred by examining the intensity variation on the road surface below it. Normally, the road surface appears brighter directly under a functioning streetlight and gradually becomes darker moving away from it until the next light source. Anomalies in this expected pattern indicate possible defects such as dimness, flickering, or complete failure.

OVERVIEW OF WORK DONE (step by step)

1. Data Collection

For this project, the primary dataset was created through real-world data collection using a mobile camera. The data was captured from the **point of view (POV) of a moving vehicle**, ensuring that the recordings closely resembled real driving scenarios where streetlights would be observed from different angles and distances.

Initially, video recordings were captured inside the Indian Institute of Science (IISc), Bangalore. This environment provided an ideal and structured setting for capturing streetlights in well-planned streets.

However, to ensure that the trained model could generalize well to diverse real-world conditions, additional data was collected from different types of road environments:

- **Mathikere residential area**
- **BEL Road**

All the video data was collected during **nighttime**, as the project focuses on assessing **streetlight performance under low-light conditions**. After collection, the videos were **split into frames** using a custom script, which allowed frame-by-frame annotation and model training. This approach resulted in a comprehensive and diverse dataset representing different lighting conditions, streetlight structures, and urban scenarios.

2. Data Annotation and Preparation

After collecting the raw video data from various locations, the next crucial step was **data annotation and preparation**, which ensures that the machine learning models receive accurate and meaningful input for training.

The collected video data was first **converted into individual frames** using a Python script. These frames were then uploaded to **RoboFlow**, an online platform widely used for dataset management and annotation in computer vision projects.

Two separate types of annotations were performed:

- **Streetlight Annotation** – Each streetlight present in the frame was manually annotated with bounding boxes. The annotations were done with high precision to ensure that the model could effectively **distinguish between actual streetlights and other light sources** such as vehicle

headlights, building lights, or illuminated signboards. This step was critical for improving the model's detection accuracy in complex urban environments.

- **Road Segmentation Annotation** – In addition to object detection, **segmentation masks** were annotated for roads. This was necessary to train a segmentation model that could identify drivable road regions, especially in low-light scenarios. Each road area in the frame was accurately outlined to provide clear ground truth for the segmentation model.

Once the annotation process was completed, the dataset was **organized and preprocessed** within RoboFlow, including:

- **Dataset splitting** into training, validation, and testing subsets to ensure unbiased performance evaluation.
- **Format conversion** to make the dataset compatible with YOLOv8 for both object detection and segmentation tasks.
- Applying **augmentation techniques** offered by RoboFlow (such as flipping or scaling, if used) to increase dataset variability and improve model robustness.

By combining precise annotations for both streetlights and roads, the prepared dataset became a **strong foundation for model training**, enabling accurate detection and segmentation under real-world nighttime conditions.

3. Model Training – Streetlight Detection

The first stage of the model training process focused on accurately detecting **streetlights in nighttime video frames**, which is a challenging task due to the presence of multiple light sources such as vehicle headlights, traffic signals, building lights, and reflective surfaces. The objective was to build a **robust and efficient object detection model** capable of distinguishing actual streetlights from other light sources under varying lighting conditions.

To achieve this, **different model architectures were initially explored**. However, after extensive experimentation, it was concluded that the most effective approach was to use **YOLOv8 for both object detection and segmentation**, as both models share a **common backbone architecture**. This unified backbone significantly improved compatibility, reduced computational complexity, and ensured seamless integration between streetlight detection and road segmentation processes. For streetlight detection specifically, the **YOLOv8 object detection model** was selected due to its **lightweight design, real-time inference capabilities, and high**

detection accuracy. The training dataset consisted of **precisely annotated streetlights**, prepared using **RoboFlow** to differentiate streetlights from other light sources in the environment. The annotated dataset was exported in a **YOLO-compatible format** and used to train the detection model.

Training Environment and Process

The model training was carried out on **Google Colab**, leveraging its GPU-accelerated environment for faster iterations. The process included the following steps:

- **Dataset Integration:**

The annotated dataset was imported from RoboFlow into Google Colab and split into training, validation, and test sets to ensure proper evaluation and avoid overfitting.

- **Model Configuration:**

A **YOLOv8 pretrained model** was used as the base, with parameters such as image size, batch size, and epochs configured based on the dataset size and GPU capacity. Using a pretrained backbone enabled the model to converge faster and achieve better accuracy with fewer samples.

- **Training Process:**

The training was performed by optimizing YOLOv8's **box loss, classification loss, and DFL loss**. After analyzing the initial results, the dataset was expanded with more diverse samples and the annotation precision was improved to handle challenging scenarios better.

- **Evaluation and Fine-tuning:**

The trained model was evaluated using **Precision, Recall, and mAP (mean Average Precision)**. Initially, the dataset collected from **IISc campus** alone was found to be too ideal, so additional data from **Mathikere residential areas, BEL Road**, and other diverse nighttime environments were collected to enhance generalization.

- **Model Export and Deployment:**

Once satisfactory performance was achieved, the trained model weights (in .pt format) were exported and integrated into a **VS Code environment**. This allowed real-time streetlight detection on video streams captured from the POV of a moving vehicle.

Model Selection Justification

While alternative models were explored, such as **DeepLab v3 for segmentation**, their performance under low-light nighttime conditions was not satisfactory. The combination of **YOLOv8 object detection and YOLOv8 segmentation models**—both sharing the **same backbone**—proved to be significantly more effective, providing higher accuracy, real-time performance, and better integration. This unified YOLOv8-based approach formed the **core detection and segmentation framework** of the system

4. Model Training – Streetlight Detection

To enhance the reliability of streetlight detection across consecutive video frames, **object tracking** was integrated into the system. Without tracking, the same streetlight could be detected multiple times in successive frames, leading to redundant detections and inconsistent intensity analysis. By introducing tracking, each streetlight is **assigned a unique and consistent ID**, enabling **stable monitoring of its corresponding road segment** over time.

Tracker Selection

For this task, the **Deep SORT (Simple Online and Realtime Tracking)** algorithm was chosen due to its efficiency and proven performance in real-time applications. Deep SORT combines **Kalman filtering** for motion prediction with **appearance feature matching**, allowing it to maintain robust object identities even in the presence of slight detection variations between frames.

Integration with YOLOv8 Detection

The output bounding boxes from the trained **YOLOv8 object detection model** were passed to the **Deep SORT tracker** in real time. For each detected streetlight:

- The bounding box coordinates, confidence scores, and class information were extracted.
- These detections were formatted and fed as inputs to the tracker.
- The tracker then assigned or updated unique IDs for each streetlight across the video frames.

As a result, each streetlight could be **tracked consistently** as the vehicle moved along the road, ensuring that **intensity values below the same streetlight were accumulated and analyzed accurately**.

Benefits of Tracking

Integrating object tracking provided several key advantages:

- **Avoiding duplicate detections:** Ensured that the same streetlight was not processed multiple times as separate detections.
- **Stable intensity analysis:** Allowed intensity values to be associated with the correct streetlight over time, improving defect detection accuracy.
- **Efficient post-processing:** Enabled the system to log streetlight-specific data (e.g., intensity patterns, defect status) using their unique IDs, which can later be mapped to their **geographical locations** if GPS data is integrated.

Visualization

During video processing, **bounding boxes and ID labels** were drawn around each tracked streetlight, making it easy to visualize the tracker's performance. The IDs remained stable even when the camera perspective changed slightly, demonstrating the effectiveness of the YOLOv8 + Deep SORT combination in real-world scenarios.

5. Intensity Measurement & Visualization

A crucial component of the defect detection system is the **measurement and analysis of light intensity on the road surface** directly beneath each detected streetlight. This step determines whether a streetlight is **functional, dim, or defective** based on the illumination pattern on the road. The analysis is carried out frame by frame, enabling real-time evaluation as the vehicle moves along the street.

Region of Interest (ROI) Selection

After **road segmentation** using the YOLOv8 segmentation model, a **Region of Interest (ROI)** was defined on the **road surface area directly beneath each detected streetlight**. This region represents the expected illuminated zone on the road if the streetlight is functioning properly.

- The **bottom portion of the road segmentation mask** was used to dynamically locate the road area under each streetlight.
- A **horizontal strip** (ROI) was defined slightly above the bottom of the segmented road to avoid noise from the vehicle headlights or edge artifacts.

This region was then isolated from the original video frame for intensity analysis.

Intensity Calculation

The selected ROI was converted to **grayscale**, and the **mean pixel intensity** of all pixels belonging to the segmented road within that region was computed. This mean intensity value serves as an **indicator of the illumination level** provided by the corresponding streetlight.

- **High intensity values** indicate a properly functioning streetlight.
- **Low intensity values** suggest that the streetlight may be **dim or not working**.
- Consistent variations between illuminated and non-illuminated sections help in distinguishing the effect of each individual streetlight.

This **automated intensity measurement** approach avoids manual inspection and allows for **continuous monitoring** along an entire stretch of road.

Visualization of Intensity Over Time

To better understand how illumination levels vary across different streetlights and locations, a **real-time intensity graph** was overlaid on the top of the processed video.

- A **scrolling line graph** was drawn for each processed frame, showing intensity variations over time as the vehicle moved.
- The X-axis represented the frame progression (i.e., spatial movement along the road), while the Y-axis represented **intensity values** (scaled typically between 0–150).
- The graph was rendered dynamically using OpenCV, allowing easy correlation between **visual streetlight performance** and **quantitative intensity data**.

Outcome

By combining precise **road segmentation**, dynamic **ROI extraction**, and real-time **intensity visualization**, the system was able to clearly distinguish between **working, dim, and defective streetlights**. This intensity analysis forms the **core decision-making layer** of the project, bridging the gap between object detection and actionable defect identification.

6. Integration and Testing

After completing the individual modules for streetlight detection, road segmentation, object tracking, and intensity measurement, the next step was to integrate them into a single unified pipeline and evaluate their performance. This

integration ensured that the system could work seamlessly in real-world scenarios and provide accurate results for defective streetlight identification.

1. Module Integration

- **Streetlight Detection:**

The YOLOv8 detection model was used to identify streetlights from the input video frames. Each detected streetlight was enclosed within a bounding box and assigned a confidence score.

- **Road Segmentation:**

A YOLOv8 segmentation model was integrated to accurately outline the road area beneath each streetlight. This segmentation provided the region of interest (ROI) where light intensity analysis could be carried out.

- **Object Tracking:**

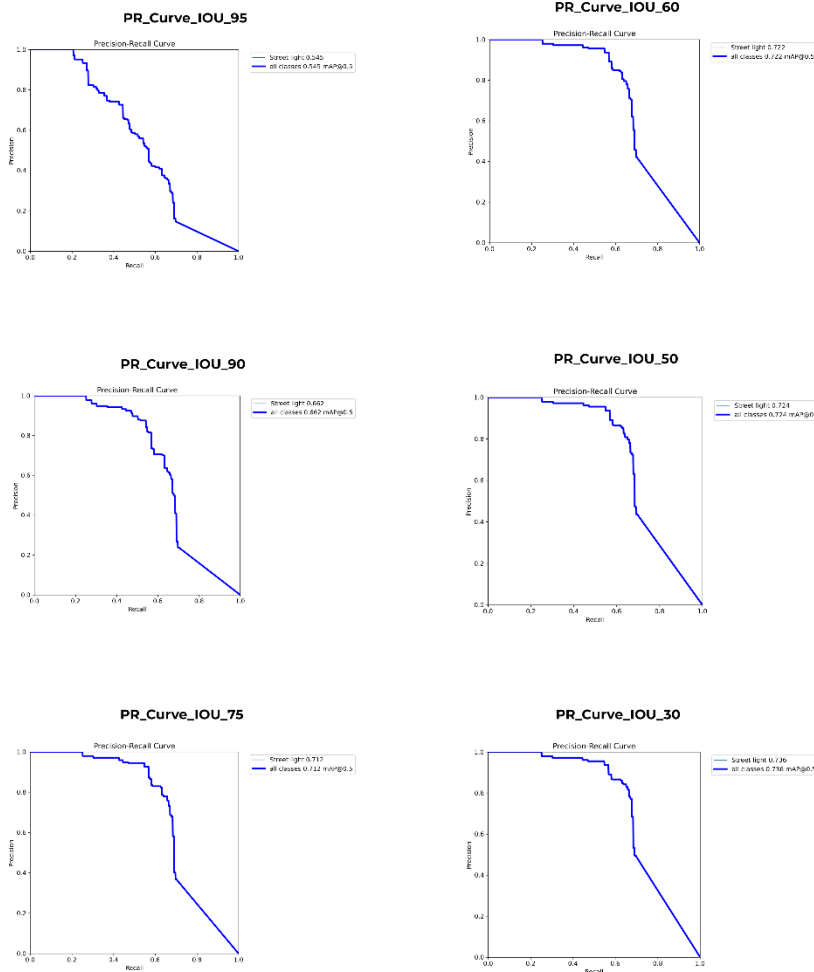
To avoid duplicate detections across consecutive frames, an object tracking module was employed. Each streetlight detected in the video stream was assigned a **unique ID**, ensuring that the same streetlight could be monitored consistently over time.

- **Intensity Measurement:**

For each detected and tracked streetlight, the intensity of the segmented road region directly beneath it was measured. This intensity analysis determined whether the streetlight was functioning correctly (bright illumination) or defective (dim or no illumination).

2. Performance Evaluation

The integrated system was evaluated using **precision-recall metrics** at different Intersection-over-Union (IoU) thresholds to determine how accurately the model could detect streetlights in various scenarios. The evaluation highlights the trade-off between precision and recall, helping to analyze how well the model generalizes across diverse environments.



3. Summary of Testing

- The system demonstrated **high accuracy in streetlight detection**, even in the presence of background noise such as vehicle headlights or building lights.
- **Road segmentation improved reliability** by focusing intensity measurement strictly within the drivable road area.

- **Tracking ensured consistency** in monitoring streetlights across frames, preventing repeated detections.
- The **intensity graph analysis** effectively differentiated between working, dim, and non-functional streetlights.

This integration and testing phase validated that the system functions as a complete pipeline capable of real-time analysis of defective streetlights in urban and rural environments.



DISCUSSION AND FUTURE SCOPE

The developed system successfully integrates multiple modules — streetlight detection, road segmentation, object tracking, and intensity measurement — to identify malfunctioning or underperforming streetlights in real time. While the implementation achieved promising results during controlled testing, several operational factors influence real-world performance and open up clear directions for future improvements.

1. Operational Analysis

During on-road deployment, various environmental and infrastructural factors were observed to affect the accuracy and robustness of the system. Key operational challenges include:

- **Other Light Sources:** Headlights from oncoming vehicles, billboard lights, and shop illuminations often introduce false intensity spikes, leading to incorrect inferences.
- **Road Surface & Environmental Variability:** Potholes, shadows, wet surfaces, fog, and glare alter road brightness readings, complicating intensity-based evaluation.
- **Segmentation Complexity:** Road intersections, crosswalks, and footpaths frequently introduce segmentation errors, especially under dynamic conditions.
- **Occlusions & Dynamic Obstacles:** Vehicles, tree branches, poles, and pedestrians occasionally block the view of streetlights or road regions, affecting both detection and intensity estimation.
- **Model and Sensor Limitations:** Camera positioning, low-light performance, and variations in streetlight design affect model generalization. Over time, **model drift** can occur, necessitating periodic retraining to maintain performance.
- **Real-Time Constraints:** Simultaneous road segmentation and object detection are computationally demanding, requiring optimized models for real-time embedded deployment.
- **Annotation and Thresholding Challenges:** Precise annotation of road regions and streetlights is time-intensive, and fixed intensity thresholds may not adapt to diverse scenarios, demanding adaptive algorithms.

These challenges underscore the importance of designing more adaptive, environment-aware algorithms for large-scale deployment.

2. Future Scope

The initial vision of the project was to develop a lightweight, real-time system that can be deployed in vehicles to autonomously monitor streetlight performance as they traverse urban and rural areas. Future developments can build upon the current framework in the following directions:

1. Embedded and Mobile Integration:

The present system can be miniaturized and deployed on **standalone embedded platforms** (e.g., NVIDIA Jetson, Raspberry Pi) or integrated into **mobile applications** used by ride-sharing or taxi drivers (e.g., Uber, Rapido).

This would enable **crowdsourced streetlight monitoring**, where vehicles act as mobile sensors.

2. **Adaptive Intensity Analysis:**

Replacing fixed thresholds with **adaptive, learning-based intensity benchmarks** that consider environmental context (e.g., weather, surface conditions) will improve reliability.

3. **Robust Segmentation and Detection:**

Incorporating **multi-modal sensor fusion** (e.g., infrared or thermal data) or advanced segmentation architectures could enhance performance in low-light and complex road scenarios.

4. **Scalable Data Pipelines and Continuous Learning:**

Establishing automated data pipelines and continuous model updates will help address **model drift**, ensuring sustained performance over time.

5. **Integration with City Infrastructure:**

Linking detection results with municipal dashboards could enable authorities to quickly identify and repair faulty streetlights, improving energy efficiency and safety.

6. **Explainability and Visualization:**

Enhanced visualization dashboards, including real-time detection overlays and historical intensity charts, can make the system more user-friendly and interpretable for operators.

CONCLUSION

This project was carried out by **Jobin Jose** under the guidance of **Dr. Prasant Misra** and **Dr. Pandarasamy Arjunan**. The work presented here is the result of continuous exploration and experimentation over multiple phases. Several approaches and methods were tested during the development cycle to arrive at the most effective solution, and only the **final methodology and results** have been included in this report for clarity.

The system integrates **streetlight detection, road segmentation, object tracking, and intensity measurement** to provide a structured framework for identifying underperforming or malfunctioning streetlights using mobile or dashboard cameras. While the current implementation demonstrates the feasibility and robustness of the proposed pipeline, it is **not the final version** of the system. There remains significant potential for **further**

enhancements and extensions, including integration into mobile applications or embedded platforms for large-scale deployment, as discussed earlier.

If the reader has any questions, requires clarification, or wishes to understand the intermediate approaches and experiments not documented in this report, they are encouraged to **contact the author, Jobin Jose**, directly. The work remains open for **future modifications and expansions**, making it a strong foundation for continued research and practical implementation.

Supporting Resources

GitHub -

<https://github.com/Jobin-555/Street-Light-Defect-Detection-Using-ML-and-CV->

Roboflow -

https://app.roboflow.com/join/eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3b3Jrc3BhY2VJZCI6InM5aEhZdEg2VDBOa09VQltaFFwUEh2aF5QjliLCJyb2xlljoib3duZXliLCJpbnZpdGVyljoiam9iaW5qb3NlMjc0N0BnbWFpbC5jb20iLCJpYXQiOiJlE3NjAxNzQ1NTd9.b9OlyVn3_Vg36mGkCV6_eR2RpYwTraxQEzYjTXU6JVc

Videos and Images -

<https://drive.google.com/drive/folders/1GLxRjZO7R5Ha0kCge428DZIBqWYxPeHj?usp=sharing>

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